

For the core users:

```
User(
  userID    INT,
  username  VARCHAR(50),
  email     VARCHAR(100),
  city      VARCHAR(50),
  createDate TIMESTAMP,
  PRIMARY KEY (userID)
);

EndUser(
  userID    INT,
  password  VARCHAR(100),
  PRIMARY KEY (userID),
  FOREIGN KEY (userID) REFERENCES User(userID)
);
```

```
Admin(
  adminID   INT,
  password  VARCHAR(100),
  PRIMARY KEY (adminID)
);
```

```
CustomerServiceRep(
  employeeID INT,
  password   VARCHAR(100),
  PRIMARY KEY (employeeID)
);
```

Features like listings, sales and bids:

```
Listing(
  listingID  INT,
  title      VARCHAR(100),
  category   VARCHAR(50),
  price      DOUBLE,
  closedTimestamp TIMESTAMP,
  userID     INT,
  PRIMARY KEY (listingID),
  FOREIGN KEY (userID) REFERENCES EndUser(userID)
);
```

```
Bid(
```

```

bidID    INT,
price    DOUBLE,
time     TIMESTAMP,
listingID INT,
userID   INT,
PRIMARY KEY (bidID),
FOREIGN KEY (listingID) REFERENCES Listing(listingID),
FOREIGN KEY (userID) REFERENCES EndUser(userID)
);

```

```

Sale(
  saleID    INT,
  listingID INT,
  amount    DOUBLE,
  dateTime  TIMESTAMP,
  PRIMARY KEY (saleID),
  FOREIGN KEY (listingID) REFERENCES Listing(listingID)
);

```

Tickets:

```

Ticket(
  ticketID INT,
  userID   INT,
  replID   INT,
  status   VARCHAR(30),
  notes    VARCHAR(500),
  PRIMARY KEY (ticketID),
  FOREIGN KEY (userID) REFERENCES User(userID),
  FOREIGN KEY (replID) REFERENCES CustomerServiceRep(employeeID)
);

```

Association relations:

```

Posts(
  userID INT,
  listingID INT,
  PRIMARY KEY (userID, listingID),
  FOREIGN KEY (userID) REFERENCES EndUser(userID),
  FOREIGN KEY (listingID) REFERENCES Listing(listingID)
);

```

```

Places(
  userID INT,

```

```
bidID INT,  
PRIMARY KEY (userID, bidID),  
FOREIGN KEY (userID) REFERENCES EndUser(userID),  
FOREIGN KEY (bidID) REFERENCES Bid(bidID)  
);
```

```
Generates(  
employeeID INT,  
saleID INT,  
PRIMARY KEY (employeeID, saleID),  
FOREIGN KEY (employeeID) REFERENCES CustomerServiceRep(employeeID),  
FOREIGN KEY (saleID) REFERENCES Sale(saleID)  
);
```

```
MakesTicket(  
userID INT,  
ticketID INT,  
PRIMARY KEY (userID, ticketID),  
FOREIGN KEY (userID) REFERENCES User(userID),  
FOREIGN KEY (ticketID) REFERENCES Ticket(ticketID)  
);
```

```
Resolves(  
employeeID INT,  
ticketID INT,  
PRIMARY KEY (employeeID, ticketID),  
FOREIGN KEY (employeeID) REFERENCES CustomerServiceRep(employeeID),  
FOREIGN KEY (ticketID) REFERENCES Ticket(ticketID)  
);
```

```
CreatesRep(  
adminID INT,  
employeeID INT,  
PRIMARY KEY (adminID, employeeID),  
FOREIGN KEY (adminID) REFERENCES Admin(adminID),  
FOREIGN KEY (employeeID) REFERENCES CustomerServiceRep(employeeID)  
);
```